

SIMATS ENGINEERING

ASSESSMENT TOOL I – Uncertainty Visualization Lab

Course: Data Handling and Visualization	Code: DSA06	CO: C05
Duration: 180 min	Total Marks: 100	Reg No : 192524225 Name: VALLAPU MOHAN KUMAR

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Uncertainty Visualization Lab

Weightage: 20%

Code of Conduct:

*I **VALLAPU MOHAN KUMAR** (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: V. Mohan Kumar

Criteria	Understanding of Uncertainty 25%	Visualization of Uncertainty 25%	Application of Visualization Principles 20%	Organization 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							

Q5							
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Uncertainty Visualization Lab – Assessment Tool

R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualization, interpret uncertainty, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

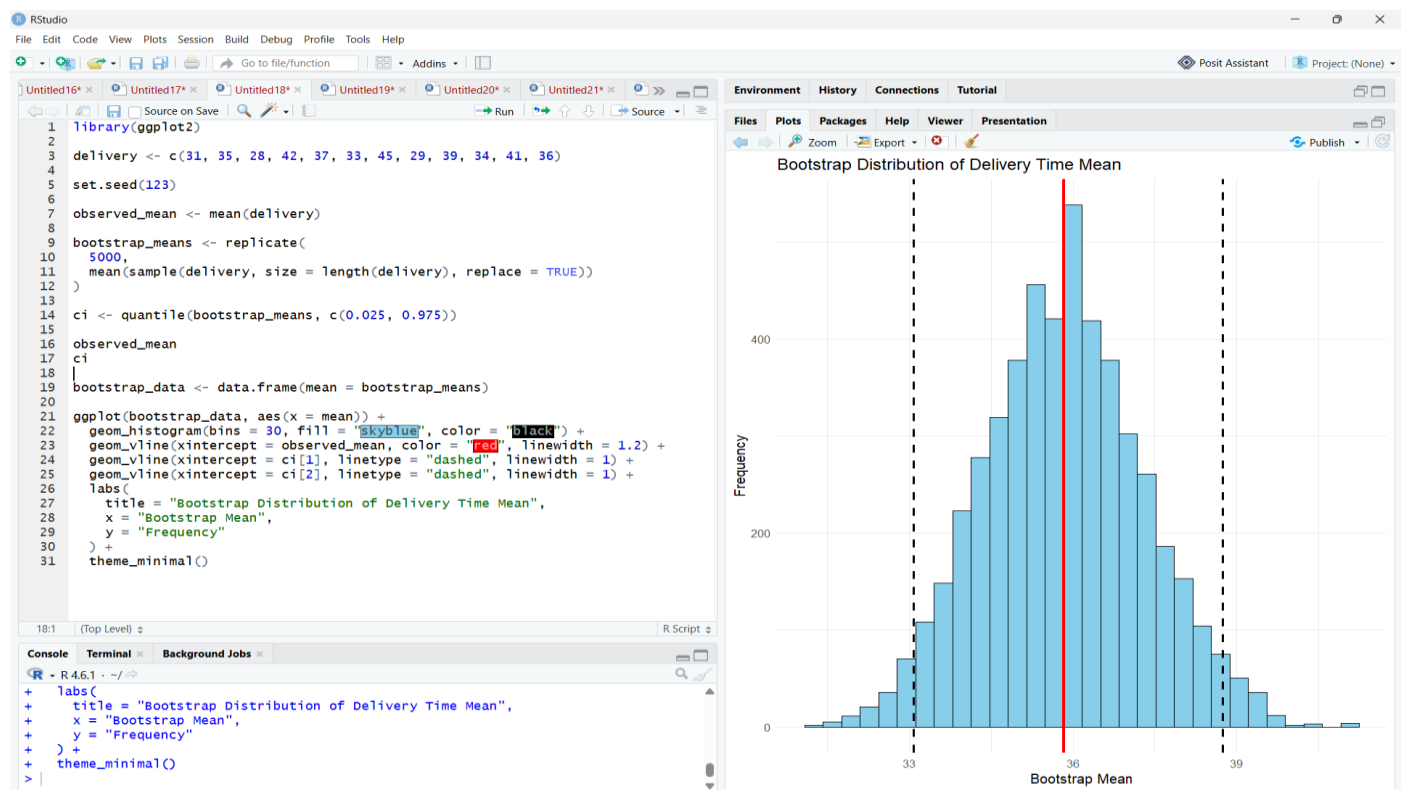
Q1. Bootstrapping and Confidence Intervals – Point Estimate

A quality-control team records the following delivery times (in minutes) for 12 randomly selected orders. Using R, calculate the sample mean and obtain a 95% bootstrap confidence interval for the mean using at least 5,000 bootstrap resamples. Visualize the bootstrap distribution and mark the observed mean.

Data: 31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36

Expected R tasks: resampling, point estimate, confidence interval, histogram/density plot, interpretation.

Answer:



Expected result

Sample mean:

$$\bar{x} = \frac{31 + 35 + 28 + 42 + 37 + 33 + 45 + 29 + 39 + 34 + 41 + 36}{12}$$

$$\bar{x} = 35.83 \text{ minutes}$$

With `set.seed(123)`, the bootstrap 95% percentile interval will be approximately:

$$33.00 \text{ to } 38.67 \text{ minutes}$$

The exact interval can vary slightly depending on the number of resamples and random seed.

Interpretation:

The observed average delivery time is approximately 35.83 minutes. The bootstrap confidence interval indicates that the population mean delivery time is reasonably estimated to lie between approximately 33 and 39 minutes. The bootstrap distribution shows the sampling uncertainty associated with estimating the mean from only 12 observations.

Q2. Uncertainty in Trend Estimation

The following data represent monthly advertising expenditure and sales. Fit a linear regression model in R to estimate the sales trend. Plot the observations with the fitted regression line and a 95% confidence band. Explain how the uncertainty changes across the range of advertising expenditure.

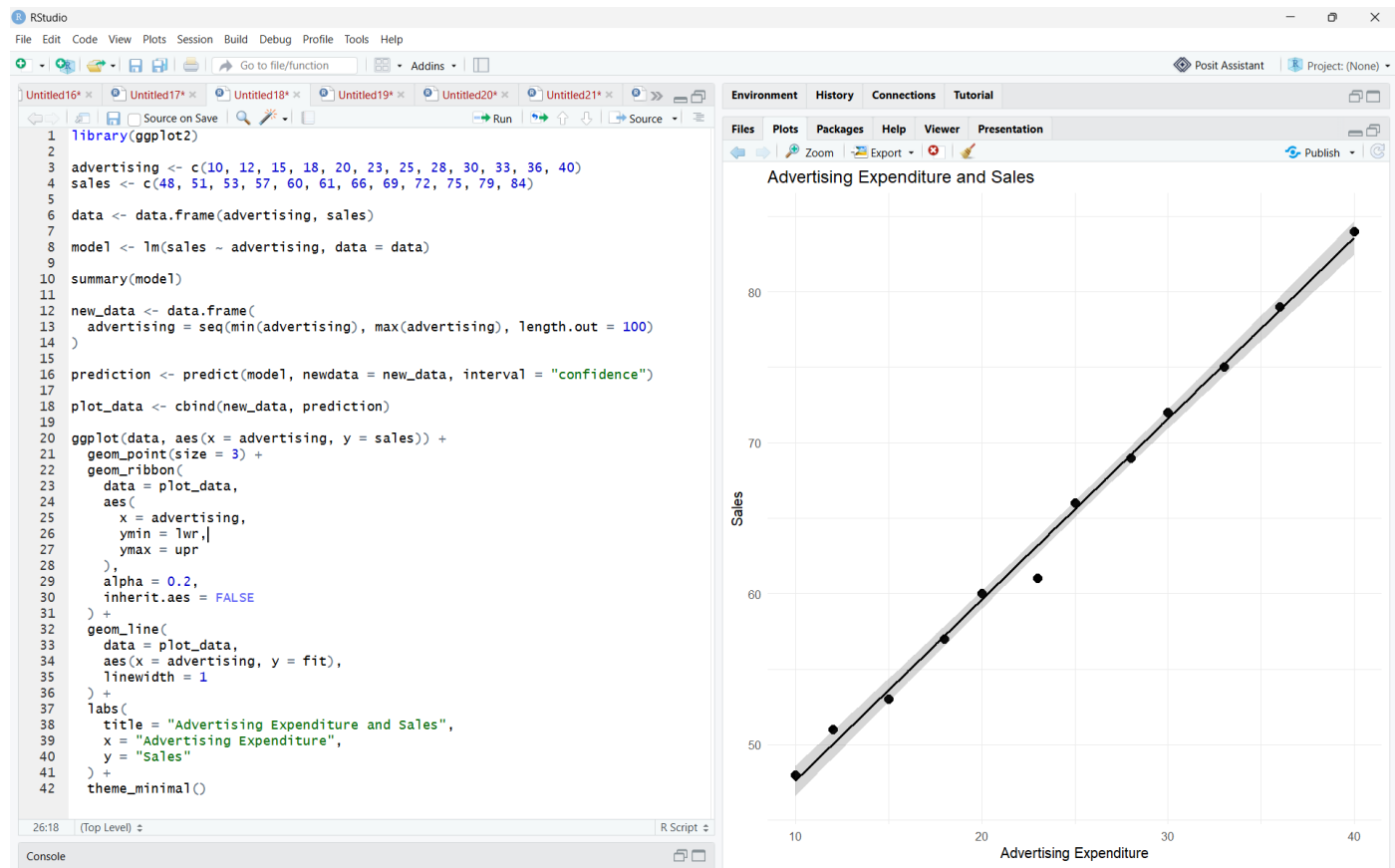
Data: Advertising = 10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40;

Sales = 48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84

Expected R tasks: `lm()`, prediction interval/confidence interval, `ggplot2` visualization, trend

interpretation.

Answer:



Regression equation

The fitted regression equation is approximately:

$$\text{Sales} = 35.616 + 1.199(\text{Advertising})$$

This means that for every one-unit increase in advertising expenditure, predicted sales increase by approximately 1.20 units.

Interpretation:

There is a clear positive relationship between advertising expenditure and

sales. As advertising expenditure increases, sales also increase.

The confidence band represents uncertainty around the estimated mean sales.

It is generally narrower near the center of the observed advertising range and becomes wider toward the boundaries because there is less information for estimating the mean at the extremes.

Q3. Overplotting, Jittering and Transparency

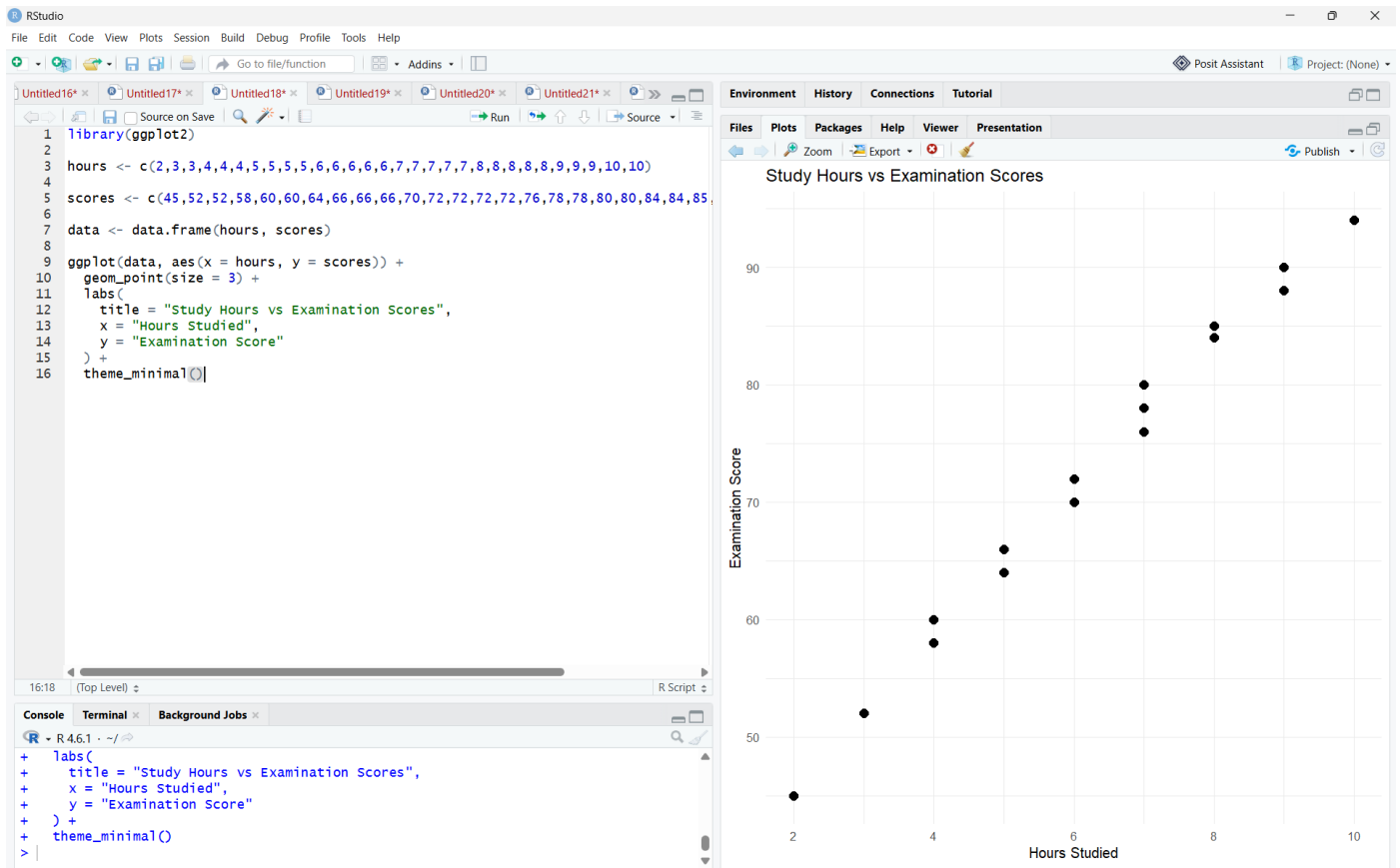
A survey records the number of hours studied and examination scores for 30 students. Several students have identical values. Create a scatterplot in R first without modification, then redesign it using partial transparency and jittering. Compare the two plots and explain how overplotting affects interpretation.

Data: Hours = 2,3,3,4,4,4,5,5,5,5,6,6,6,6,6,7,7,7,7,7,8,8,8,8,9,9,10,10;

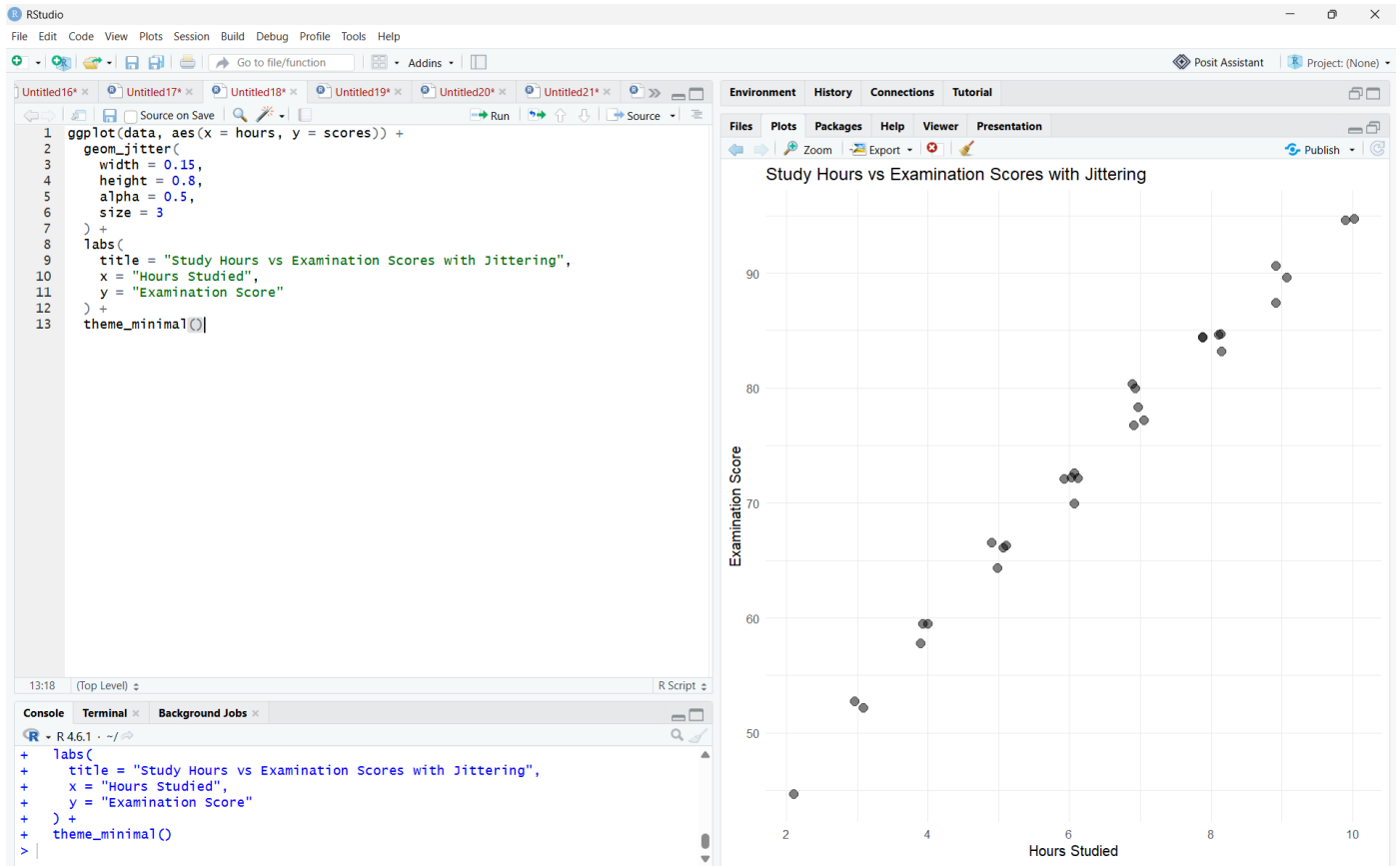
Scores = 45,52,52,58,60,60,64,66,66,66,66,70,72,72,72,72,76,78,78,80,80,84,84,85,85,85,88,90,90,94,94

Expected R tasks: `geom_point()`, `alpha`, `position_jitter()`, comparison of overplotting.

Answer:



Improved plot using transparency and jittering



Interpretation

The first scatterplot suffers from overplotting because several students have identical study hours and examination scores. Some points overlap and therefore the number of observations at particular locations may be underestimated.

Jittering slightly moves the points while retaining their approximate original locations. Transparency makes overlapping observations darker, allowing areas with many observations to become more visible.

The second visualization communicates the distribution of observations more clearly than the original scatterplot.

Q4. 2D Histogram and Density Estimation

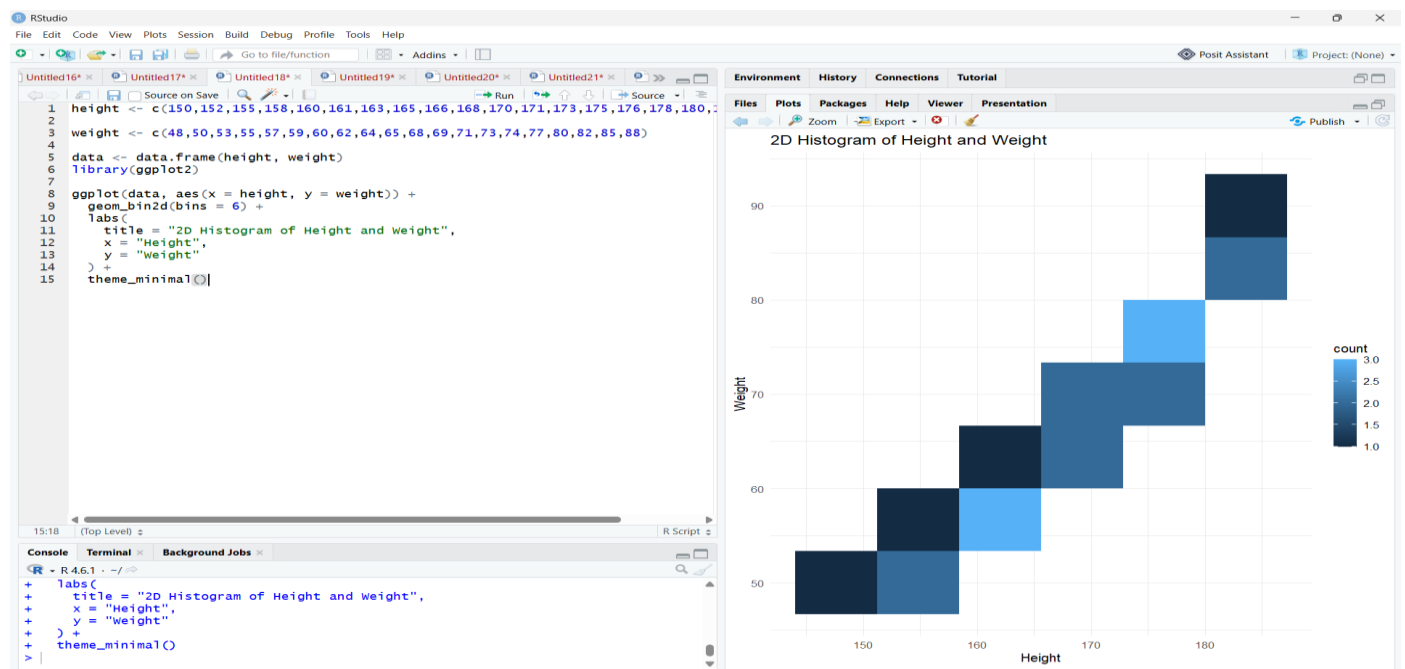
A researcher measures height and weight for 20 participants. Construct a 2D histogram and a 2D density visualization in R. Use the plots to identify regions with high observation density. State which visualization communicates concentration more clearly and justify your choice.

*Data: Height = 150,152,155,158,160,161,163,165,166,168,170,171,173,175,176,178,180,182,184,186;
Weight = 48,50,53,55,57,59,60,62,64,65,68,69,71,73,74,77,80,82,85,88*

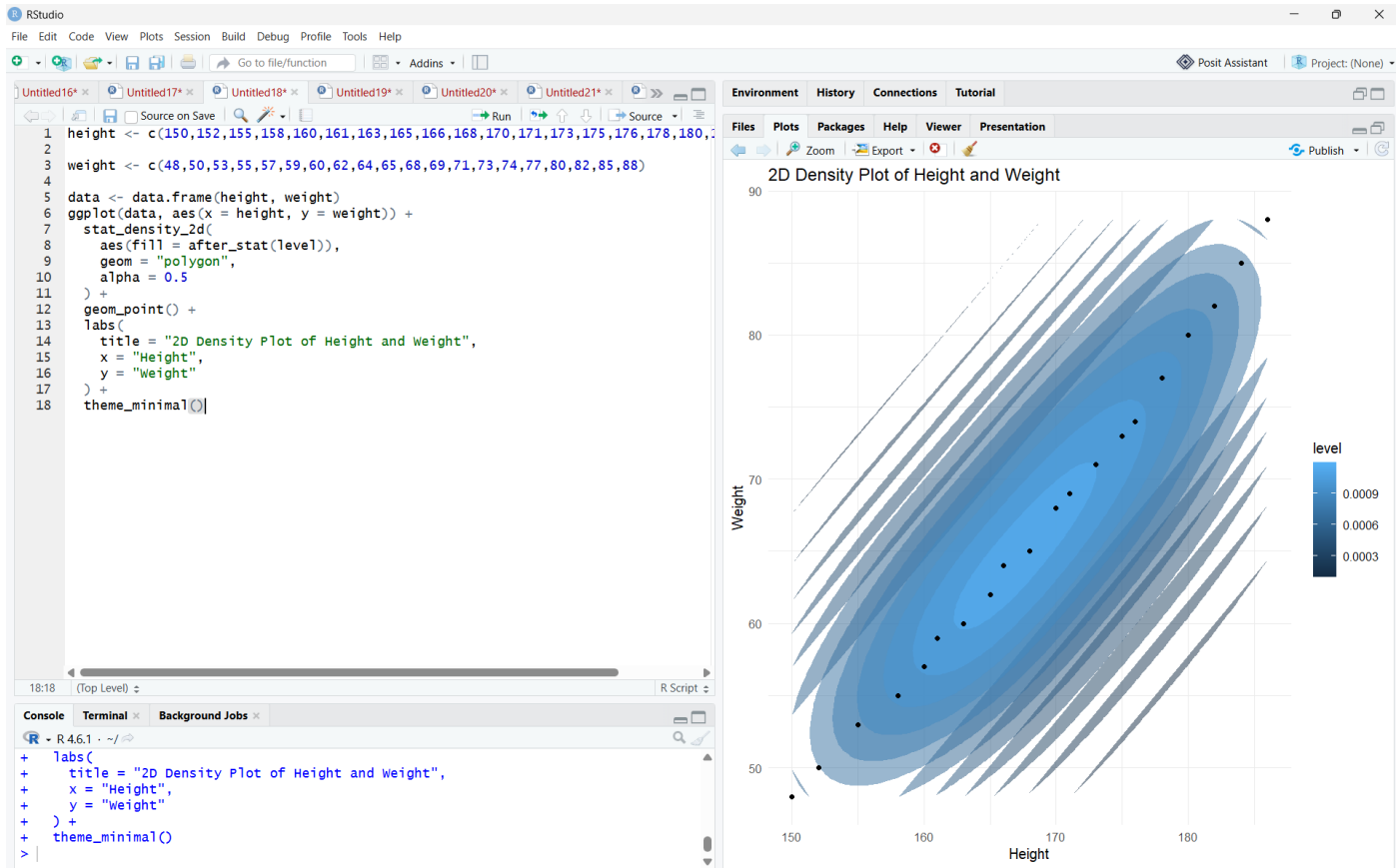
Expected R tasks: 2D binning, density estimation, contour/density layers, interpretation of concentration.

Answer:

2D Histogram



2D Density Plot



Interpretation

The observations show a clear positive relationship between height and weight.

As height increases, weight generally increases.

The highest concentration of observations occurs around the middle portion of the height-weight range, approximately around heights of 160–175 cm and weights of 57–74 kg.

The 2D histogram divides the data into rectangular bins and represents the number of observations within each bin.

The 2D density plot provides a smoother representation of concentration.

Therefore, the 2D density visualization communicates the concentration more

clearly, because it shows the overall structure and high-density regions without depending as strongly on arbitrary bin boundaries.

Q5. Color Scales, Accessibility and Misleading Visualization

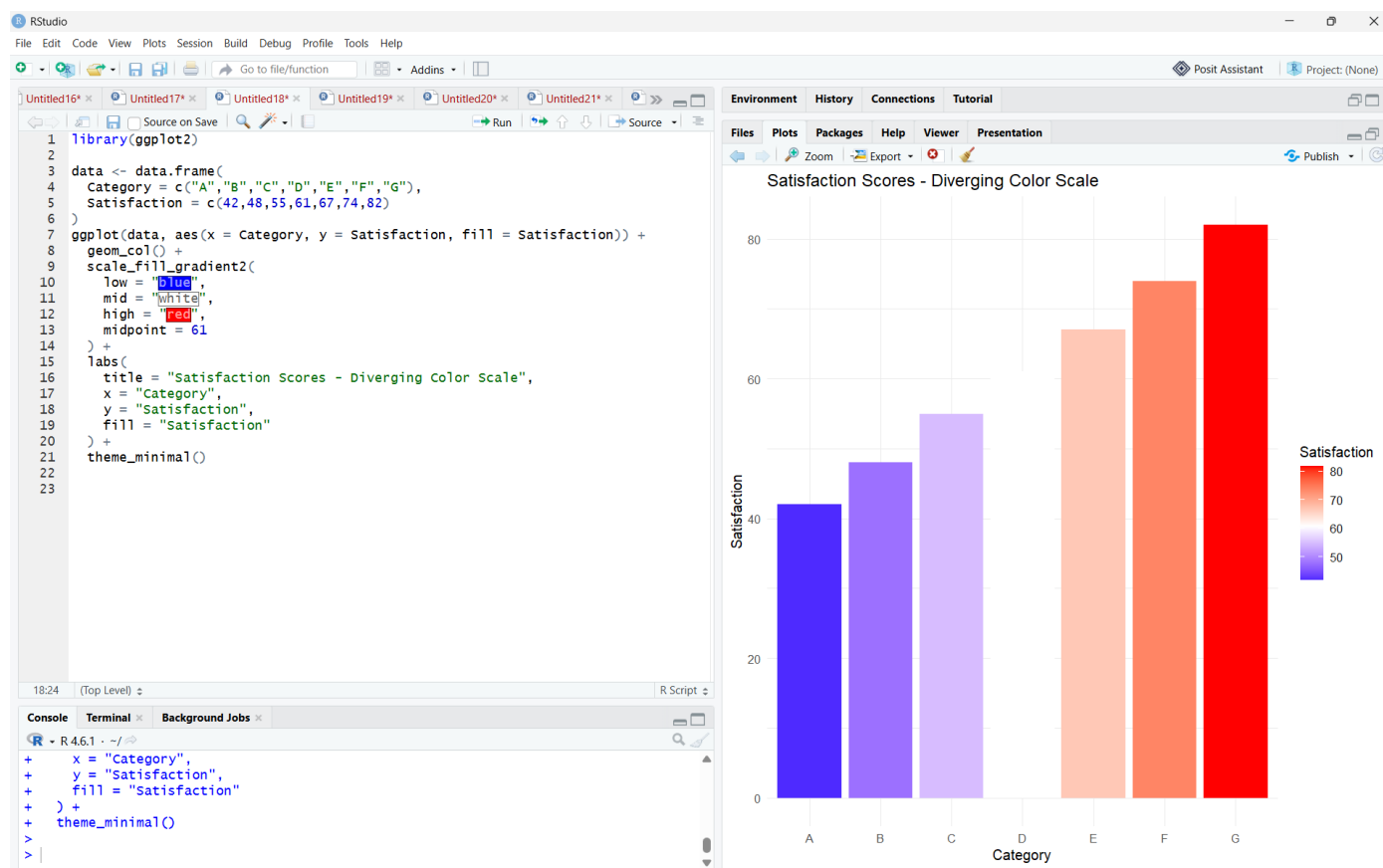
The following category-wise satisfaction scores must be visualized in R. Create two plots: (1) a plot using a nonmonotonic/diverging color scheme that could mislead interpretation, and (2) a redesigned plot using an ordered, color-vision-deficiency-friendly sequential scale. Explain why the second design is more effective and identify the visualization pitfall in the first design.

Data: Category = A,B,C,D,E,F,G;

Satisfaction = 42,48,55,61,67,74,82

Expected R tasks: color mapping, scale selection, accessibility-aware redesign, critique of misleading color use.

Answer:



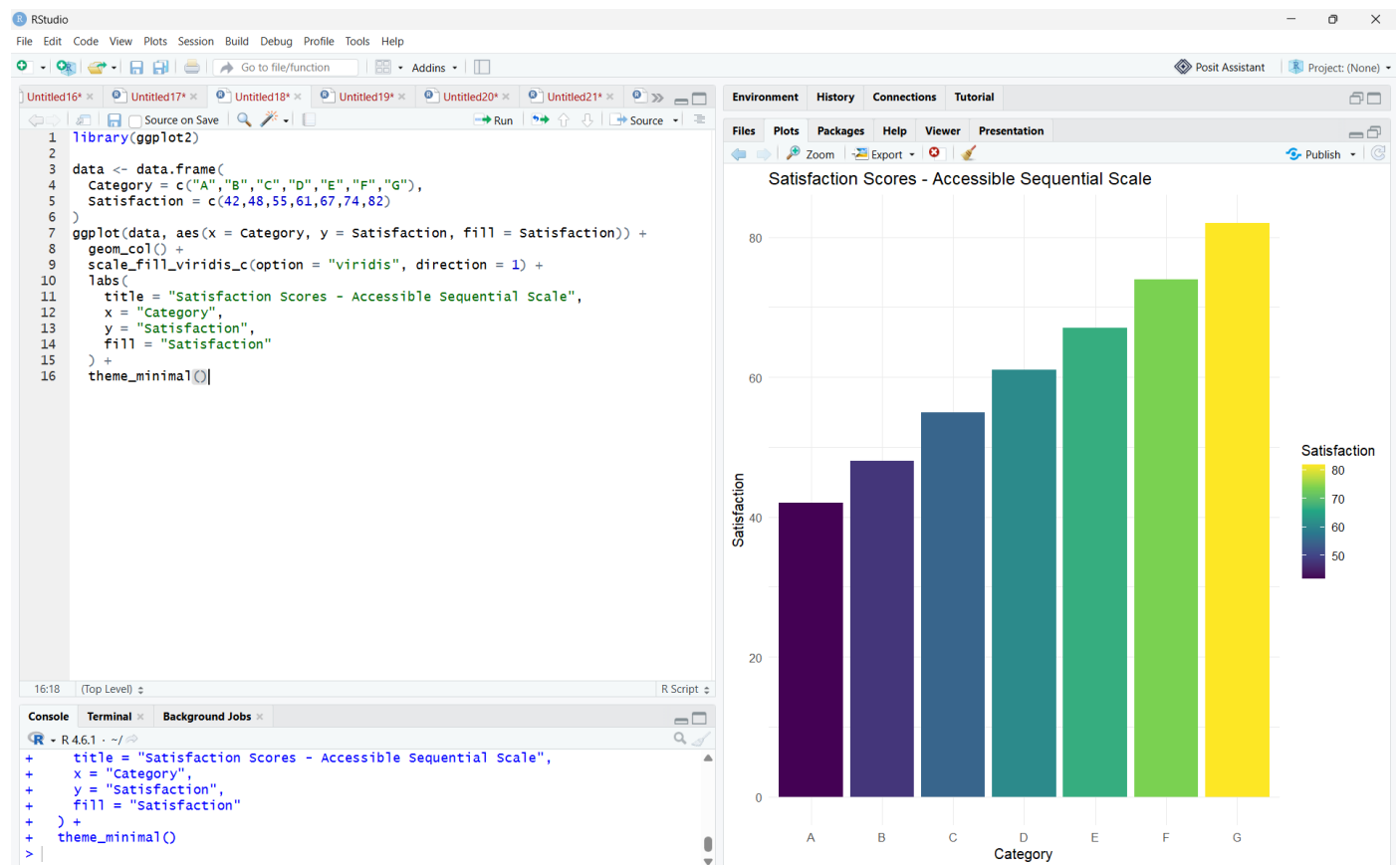
Problem with Plot 1

The diverging scale suggests that values around the midpoint are fundamentally different from values above and below it.

However, satisfaction is an ordered numerical variable. The values progress naturally from low to high:

$$42 < 48 < 55 < 61 < 67 < 74 < 82$$

A diverging color scheme can therefore introduce a misleading visual emphasis and may imply a meaningful central threshold that does not actually exist.



Interpretation

The second plot uses an **ordered sequential color scale**, which is more appropriate because satisfaction scores have a natural low-to-high ordering.

The viridis scale is designed to provide good perceptual ordering and improved accessibility for people with different forms of color-vision deficiency.

Therefore, the second visualization communicates the numerical progression more accurately and avoids the misleading emphasis created by the diverging scale

Suggested R packages: ggplot2, dplyr, boot (or base R).

Assessment focus: Correct R code, appropriate uncertainty visualization, interpretation, and identification of misleading design choices.